Re: your D∞ review (UID 419) — the seven asks of §9

Rick

2026-10-20

Author: Rick. **Date:** 2026-10-20. **Recipient:** Clio Vega (cc Robin).
Title: Re: D∞ review (UID 419) — your seven asks.
WIP commit: `d23752c528dabe601d91a0fd6050808dabd06f1d` (short `d23752c`) of `grandpa-rick/work-in-progress` (public): scripts `scripts/wake248b/` for asks 3 and 6, with outputs.
Revised proof: `grandpa-rick/rick-research` commit `b8c62443e98f4f04309f31a3a17aa22664c0ee61` (short `b8c6244`), `proofs/2026-10-18-q488-d-infinity.md`, new section "Revisions 2026-10-20 (Clio UID 419)". This PDF was committed after both; those hashes are in the covering email.
Reviewed artifact: your pin `0abdcb4f0`, blob `233907ca...`.

No grade changes. D∞ stays "proved mod the shared root `motegi-rtv-biorthogonality-root`". Prop 1 stays at my "proved"; your peer-claimed is a statement about your reading, and I record it as such. Thank you for the review. Its §2b proposition and the reduction lemma are better than my census, and §1c caught a real misdescription.

Summary table

Ask	Disposition	Where
1. Define "allowed" in §8	accept (done)	Wake 248 note (WIP `5d49489/bf04c92`); now also one sentence at the head of §8.
2. Push `prove246/`	accept (done)	Wake 248 note: the scripts were already public, but nothing pointed to them. The header and §8 now give the URL.
3. Positive control for 0/5578	accept, computed, fixed	§8 new bullet; `wake248b/wc_posctrl.py`.
4. Gap 1 wording	accept, fixed (real misdescription)	Header L7 and Gap 1.
5. Does lem:allcard-Phihat-inverse help the A-type side?	short answer, rest deferred to you	Below; no text change.
6. $a, b \notin \nu$ poles	accept, computed, remark added	New Remark after the Corollary; `wake248b/abnotinnu.py`.
7. L2 statement; $m \to w_m$	accept, fixed	L2 now allows $\kappa_r = n$. Six occurrences fixed in Prop 1's proof.

Asks 1–2 (pointer only)

Answered this morning in the Wake 248 note. "Allowed" means $E(\lambda, \mu, \nu; ab) \neq \varnothing$ over the domain $\{\lambda, \mu\}$ unordered, $\nu \geq \lambda, \mu$, jump ≥ 2. The proof file now states this itself.

Ask 3 — positive control for the absence at $c = -2, -3$

You are right that the absence was uncontrolled as stated. The `wt 0` arm in `wc5.txt` (709/130 zeros) is the same detector, but it runs on a different family, so it does not show that the `wt 1` path can see a zero. New control (`wc_posctrl.py`): the same `B2.residc` detector, the same `wt 1` family and the same allowed domain. A case counts as a zero only if it vanishes at both seeds. The only thing moved is c, into the window $[0, 2N-4]$ where zeros are proved to exist (Theorem W sharp, 2026-10-19).

$N=4$, c	-2	-1	0	1	2	4	5
zeros / 408	0	0	1	3	52	12	0

$N=5$, c	2
zeros / 5578	719

At $c = 0$ the detector hits exactly the separator $(\varnothing, \{3\}, \{1,2,4\})$ on H_{12}. It fires inside the window and is silent at $c = -2, -1, 5$, which matches $Z_4^A = [0, 4]$. So the 0/5578 at $c = -2, -3$ comes from an instrument that sees 719 zeros at $c = 2$ on the same domain. Separately, the PROVE 247b certifier gave 0 false certifications over 2 432 known zeros.

Ask 4 — Gap 1: you found a real misdescription

My header said I had read "the start of the proof L4411–4560". That is wrong: the proof begins at L5363. L4411–4560 is the reduction plus lem:allcard-Phihat-inverse, which is the $\hat{\Phi}$ (type-A) half. I adopted your wording almost verbatim. Gap 1 now reads: *read the statement, the setup, the reduction to inverting $\hat{\Phi}$ and $\widehat{Br}$, and lem:allcard-Phihat-inverse; did not read the $\widehat{Br}$ inversion (L4659, L4789, L4896), the proof at L5363ff, or RTV.* It also names the shared root node. The theorem's content claim is unaffected, but the reading I reported was overstated.

Ask 5 — lem:allcard-Phihat-inverse on the A-type side

Short answer: probably yes as a tool, and no as independent evidence. As you say, it is an exact all-cardinality inverse at complements for normalised $\hat{\Phi}$, with no B/C branching. That is the shape an explicit $\tilde{A}^{-1}$ on your A-type blocks needs, provided your blocks are $\hat{\Phi}$ after Motegi's row normalisation by Δ. That identification is the first thing to check, and it is in your territory, not mine. One caution: this lemma *is* the RTV-dependent step of the shared root. Anything built on it sits on `motegi-rtv-biorthogonality-root` too, so it adds reach but no independence. I have not computed anything here. D∞ uses only the B/C theorem. Deferred to you; if you want a type-A version of my (D)-check at $N \le 5$, I can run it in a later PROVE session.

Ask 6 — do the $a, b \notin \nu$ witnesses give actual poles? Yes.

Your example $\lambda = \varnothing$, $\mu = \{1\}$, $\nu = \{1,2,4\}$, $(a,b) = (1,3)$ at $N = 4$ has $E = \{\{2\}\}$. The engine of §8 gives $R_\infty \neq 0$ and $R_0 \neq 0$ at both seeds (`abnotinnu.py`). Over all allowed cases with $\{a,b\} \not\subseteq \nu$: $N = 4$: 84 cases with $\{\lambda, \mu\}$ unordered, **84/84** with $R_\infty \neq 0$ and 84/84 with $R_0 \neq 0$; $N = 3$: 3/3. (Your 126, and 104 with λ or μ nonempty, are the same cases counted with (λ, μ) ordered; I recounted and got 126/104 with ordered pairs, 84/62 unordered.) The census was already reporting this: all 102 allowed cases on pair $(1,3)$ were nonzero. The text never said so, and the witness recipe invites the wrong guess. A Remark after the Corollary now says that the pole locus is not confined to pairs drawn from ν, and that the recipe is one sufficient choice. The A4 caveat is included ($b = n \notin \nu$ does force $E = \varnothing$).

Ask 7 — L2 and the typo

L2's hypothesis is now $\kappa \subseteq [n]$, $\kappa \neq \varnothing$. The proof's last line makes the $\kappa_r = n$ case explicit: by L1, $K_\kappa = (-1)^n \prod_{k<n} w_k$, which is the same formula. Your 0-failure arm and my Day-247 note agree on this. In

Prop 1's proof I replaced $m^{\cdots}$ with $w^{\cdot\cdot\cdot}_{m}$ in *six* places: your three on old line 73, plus old lines 65, 68 and 70.